

KSA Office IT Support & Infrastructure Management

Phase 07: IT Support Operations

Afaq Business Solutions Co.

Access & Permissions Troubleshooting SOP

1. Purpose

Provide a structured method for investigating Linux access and permission incidents, applying controlled corrections, verifying access, and documenting the result.

2. Workflow

- Document the reported access problem and create a ticket.
- Inspect file ownership and permissions with **ls -l** and **stat**.
- Inspect relevant directory permissions with **ls -ld**.
- Review effective group membership with **id**.
- Identify the permission or group mismatch.
- Apply the smallest authorized correction.
- Verify ownership, permissions, group membership, and access.
- Document findings and resolution.
- Close the ticket after successful verification.

3. Key Commands

- **ls -l** - inspect file permissions and ownership.
- **ls -ld** - inspect directory permissions.
- **stat** - inspect detailed file metadata.
- **id** - review effective user and group membership.
- **chgrp** - change group ownership when authorized.
- **chmod** - change permission bits in a controlled manner.
- **newgrp** - start a shell using a selected effective group.

4. Controlled Remediation

- Use a dedicated lab resource.
- Assign the resource to the appropriate support group.

- Grant only the required permission, such as group read access.
- Do not use broad permissions such as 777 as a troubleshooting shortcut.

5. Verification

- Recheck group ownership and permission bits.
- Verify effective group membership.
- Test access to the controlled resource.
- Document successful verification before closure.

6. Security

- Use sanitized training resources.
- Do not expose credentials, tokens, real sensitive data, or unrelated system information.
- Capture only evidence necessary to support the incident.

7. Best Practices

- Inspect before changing.
- Follow least privilege.
- Make the smallest authorized change.
- Verify after remediation.
- Maintain a clear audit trail.